

We start with a one-step method
with order of accuracy $p \geq 2$:

$$u_1 - u_0 = \mathcal{O}(h^{p+1})$$

$$\text{Invariant: } \frac{d}{dt} I(u) = I'(u)f(u) = 0. \quad \forall u$$

$$\text{We want: } I(u_1) = I(u_0)$$

$$\text{Relaxation: } \hat{u}_1 = u_0 + \gamma(u_1 - u_0)$$

$$\text{Solve: } I(\hat{u}_1) = I(u_0)$$

$$\text{and set } \hat{t}_1 = \gamma h$$

Accuracy

$$u_0 = u(h) - hu'(h) + \mathcal{O}(h^2)$$

$$u_1 = u(h) + \mathcal{O}(h^{p+1})$$

$$\hat{u}_1 = u_1 + (\gamma-1)(u_1 - u_0)$$

$$= u(h) + \mathcal{O}(h^{p+1}) + (\gamma-1)(u_1 - u_0)$$

$$= u(h) + (\gamma-1)hu'(h) + \mathcal{O}(h^{p+1}) + \mathcal{O}((\gamma-1)h^2)$$

Meanwhile

$$u(\gamma h) = u(h + (\gamma-1)h)$$

$$= u(h) + (\gamma-1)hu'(h) + \mathcal{O}((\gamma-1)^2 h^2)$$

Subtract:

$$\begin{aligned}\hat{u}_1 - u(\gamma h) &= \mathcal{O}(h^{p+1}) + \mathcal{O}((\gamma-1)^2 h^2) \\ &\quad + \mathcal{O}((\gamma-1)h^2)\end{aligned}$$

If $\gamma-1 = \mathcal{O}(h^{p-1})$ then the one-step error is $\mathcal{O}(h^{p+1})$.

Size of δ -1

Consider the residual:

$$r(\gamma) = I(\hat{u}_1) - I(u_0)$$

$$= I(u_0 + \gamma(u_1 - u_0)) - I(u_0)$$

$$= \cancel{I(u_0)} + \gamma I'(u_0)(u_1 - u_0) + \mathcal{O}(h^2) - \cancel{I(u_0)}$$

$$= \gamma I'(u_0)(u_1 - u_0) + \mathcal{O}(h^2)$$

So

$$r'(1) = I'(u_0)(u_1 - u_0) + \mathcal{O}(h^2)$$

$$= I'(u_0)(\cancel{u_0} + hf(u_0) + \mathcal{O}(h^2) - \cancel{u_0}) + \mathcal{O}(h^2)$$

$$= \mathcal{O}(h^2)$$

Now $r(l) = I(u_1) - I(u_0) = \mathcal{O}(h^{p+1})$

This suggests that there is a root γ satisfying

$$1 - \gamma = \mathcal{O}\left(\frac{h^{p+1}}{h^2}\right) = \mathcal{O}(h^{p-1})$$

This can be proven rigorously ^{for small enough h} if $I(u)$ is convex or in general if

$$I'(u_1) \cdot \frac{u_1 - u_0}{\|u_1 - u_0\|} = ch + \mathcal{O}(h^2) \quad \text{with } c \neq 0$$

(implicit fcn. theorem)

Dissipative systems

We say $u'(t) = f(u)$ is dissipative w.r.t. functional $\eta(u)$ if $\eta: \mathbb{R}^n \rightarrow \mathbb{R}$

$$\frac{d}{dt} \eta = \eta'(u) f(u) \leq 0. \quad \forall u$$

For example, η may represent energy or entropy.

In this case we would like to enforce discrete dissipation:

$$\eta(u_1) \leq \eta(u_0)$$

This can also be done via relaxation.

In practice, we want to enforce not only the correct sign of the derivative but also accurate evolution.

With a RK method,

We require also $b_j \geq 0$.

$$\eta_{\text{new}} = \eta(u_0) + h \sum_j b_j \eta'(y_j) f(y_j)$$

$$\rightarrow \int_0^h \frac{d}{dt} \eta(s) ds$$

gives an estimate of the correct entropy at the next step.

We then set (as before)

$$\hat{u}_1 = u_0 + \gamma h \sum_j b_j f(y_j)$$

and solve for γ such that

$$\eta(\hat{u}_1) = \eta_{\text{new}} \leq \eta(u_0)$$

This approach has been used with great success in CFD to enforce the 2nd law of thermodynamics without resorting to implicit methods.

$$b_i a_{ij} + b_j a_{ji} + b_i b_j \leq 0 \quad \forall i, j$$

Other approaches:

- Discrete gradients
- Averaged vector-field method
(continuous-stage RK methods)

Homework: Implement projection and relaxation for one of the combinations of problem + method. Test how this affects the numerical error over long simulation times.

For the reference solution, you can code the exact solution or use a numerical solution that is much more accurate than the one you're testing.